

Duration: 3 Hours

Max. Marks: 100

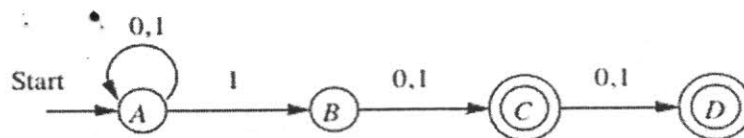
Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**

Unit – I

1. a) Define the following with suitable examples.
i) Alphabet ii) String iii) Powers of alphabet iv) length of a string
- b) For $\Sigma = \{a, b\}$, design dfa's that accept the following languages
i) all strings with even number of a's and b's
ii) all strings without aba as substring
iii) all strings with number of a's divisible by 5 and number of b's divisible by 3
2. a) Design DFA to accept decimal numbers divisible by 4
b) Construct DFA equivalent to the following automaton

	ϵ	a	b	c
$\rightarrow p$	$\emptyset$	$\{p\}$	$\{q\}$	$\{r\}$
q	$\{p\}$	$\{q\}$	$\{r\}$	$\emptyset$
*r	$\{q\}$	$\{r\}$	$\emptyset$	$\{p\}$

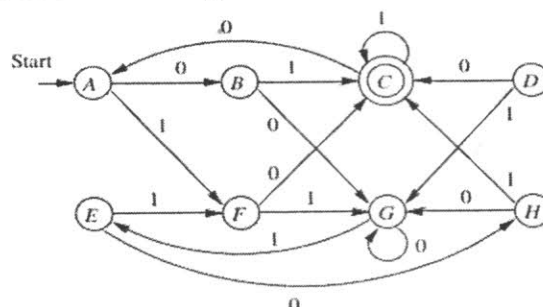
- c) Differentiate between deterministic and non deterministic automata
3. a) Find a regular expression corresponding to the language of all strings over the alphabet $\{a, b\}$
i) do not end with 'ab'
ii) at least one 'a' and at least one 'b'
iii) starting with 'a' and ending with 'b'
- b) Convert the following DFA to Regular Expressions by Eliminating State



- c) Prove that $L = \{0^n 1^n 2^n, n \geq 1\}$ is not regular.

Unit – II

4. a) Show that regular languages are closed under intersection.
b) Let G be the grammar $S \rightarrow aB \mid bA, A \rightarrow a \mid aS \mid bAA, B \rightarrow b \mid bS \mid aBB$. Find the leftmost and rightmost derivation for the string aaabbabbba
c) Minimise the following DFA



8 L4 3 2

5. a) Design context-free grammars for the following languages

i) $L = \{ w : w \in \{a, b\}^*, n_a(w) = n_b(w) \}$

ii) The set of all strings with twice as many a's as b's.

iii) $L = \{wcw^R \mid w \in (a, b)^*\}$

06 L3 3 1
08 L4 4 3

- b) Design a NPDA for accepting the language $L = \{a^n b^n c^n \mid n \geq 1\}$

- c) Check whether the following grammar is ambiguous or not for string

$w = aabbccdd$

$S \rightarrow AB / C$

$A \rightarrow aAb / ab$

$B \rightarrow cBd / cd$

$C \rightarrow aCd / aDd$

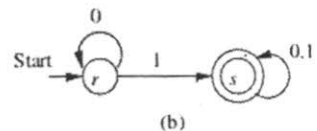
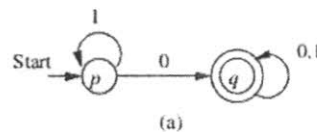
$D \rightarrow bDc / bc$

06 L3 3 2
06 L2 3 2

6. a) Explain the language accepted by PDA.

- b) Let L_1 and L_2 be the languages accepted by the following DFA's.

Find $L_1 \cap L_2$ and $L_1 - L_2$



08 L4 4 3

- c) Design a PDA that accepts strings with the same number of 0's and 1's.

06 L3 4 2

Unit – III

7. a) Remove all unit-productions, all useless productions, and all ϵ -productions from the grammar.

$S \rightarrow aA \mid aBB$ $A \rightarrow aaA \mid \epsilon$ $B \rightarrow bB \mid bbC$ $C \rightarrow B$

06 L3 5 2

- b) Show the ID's of the Turing machine if the input tape contains:

a) 00.

b) 000111.

c) 00111

State	Symbol				
	0	1	X	Y	B
q_0	(q_1, X, R)	—	—	(q_3, Y, R)	—
q_1	$(q_1, 0, R)$	(q_2, Y, L)	—	(q_1, Y, R)	—
q_2	$(q_2, 0, L)$	—	(q_0, X, R)	(q_2, Y, L)	—
q_3	—	—	—	(q_3, Y, R)	(q_4, B, R)
q_4	—	—	—	—	—

08 L4 5 3
06 L2 5 2

- c) Write a short note on Multitape Turing Machine.

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|-------|---|----|----|---|---|
| 8. a) | Design a turing machine to accept the language $L = \{a^n b^n c^n \mid n \geq 1\}$ | 08 | L3 | 5 | 2 |
| b) | Convert the following grammar in to GNF.
$S \rightarrow ABb \mid a, A \rightarrow abA \mid b, B \rightarrow aAb$ | 06 | L3 | 5 | 2 |
| c) | Explain Chomsky Hierarchy.with suitable examples. | 06 | L4 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
